

Hongdao Meng

mycrofthd@gmail.com | github.com | linkedin.com | (718) 3063737 | New York City, NY

EDUCATION

New York University Master of Science in Computer Science GPA: 3.83	Sep. 2024 - May 2026 (Expected) New York, NY
Beijing University of Technology Bachelor of Engineering in Information Security GPA: 4.00	Sep. 2020 - Jul. 2024 Beijing, CN

PROFESSIONAL & RESEARCH EXPERIENCE

Software Engineer Intern @ TikTok, Recommendation System Infra, Seattle • Developed a Python-based DAG-DSL migration toolkit (leveraging Protobuf and regex) to auto-convert legacy configuration files of the Bytedance Feature Service (BFS), a scalable feature extraction service powering TikTok’s recommendation system, into standardized DSL definitions, boosting operator migration throughput by 85%. • Designed and implemented a Blade-based C++ operator migration pipeline for BFS, streamlining the workflow to improve operator migration speed by 83.3%, and personally migrated 20 operators. • Architected a one-stop workflow for User Data Accessor covering service build, auto generate python operators, DSL compilation, and RPC request, enabling collaboration among dozens of engineers and accelerating development by 73.3%.	May 2025 - Present
Machine Learning Engineer Intern @ C2SMARTER Center, New York • Led 6-member team to develop RAG-based chatbot system using LangChain and Flask-React, achieving 24.3% accuracy improvement on MS MARCO dataset (F1=0.86) with 33.7% faster response latency through query optimization • Built hybrid retrieval framework with MilvusDB vector database and BGE-M3 embeddings, improving search relevance by 21.7% and boosting query performance 25.6% through unified re-ranking architecture • Implemented a Docker-based data pipeline with MongoDB on AWS EC2, reducing deployment setup time by 15.7% and deploying CI/CD pipelines with Jenkins to ensure high availability.	Feb. 2025 - Present
Machine Learning Engineer and Founder @ DeepFake Detection Startup, New York • Led 5-member team to develop core modules of deepfake detection web platform using React and TypeScript for seamless real time interaction, enabling 1,200+ concurrent users and reducing client-side rendering latency by 21.3%. • Fine-tuned Vision Transformer and VGG16 models from Hugging Face for image and audio DeepFake detection, achieving 91.2% and 88.1% accuracy, respectively. Constructed two custom datasets from the MP4 FaceForensics corpus by extracting video frames with OpenCV and audio files with FFmpeg , and deployed both models on AWS EC2. • Built a real-time communication layer using Django and WebSocket for robust middleware communication, reducing task completion time by 25.6% with <180ms P95 latency, and deployed backend services on Kubernetes with AWS ELB load balancing and HPA policies, achieving 99.5% availability under 5k RPM. • Optimized PostgreSQL query execution through composite index tuning, reducing average response time by 18% (320ms→262ms).	Sep. 2024 - Dec. 2024
Machine Learning Engineer Intern @ Data Mining & Security Lab, Beijing • Led research on Federated Multi-View Multi-Label Machine Learning. Published first-author paper in <i>IEEE Transactions on Big Data</i> 2025: “ Federated Multi-View Multi-Label Classification ” (DOI:10.1109/TBDDATA.2024.3522812). • Devised solutions to complex data privacy challenges by proposing and developing the FMVML framework, a federated learning method enabling cross-view feature fusion and multi-label semantic classification, which outperformed all state-of-the-art methods, improving Average Precision by 8.3% and lowering One Error by 14%. • Utilized Python/PyTorch for model development and Matlab for signal processing; implemented data pipelines with Pandas/Scikit-Learn ; produced publication-ready documents with LATEX.	Sep. 2022 - Jul. 2024

SKILLS

Languages: C/C++, Java, Python, Go, SQL, JavaScript, HTML/CSS, Shell, PHP, LATEX
Machine Learning: PyTorch, LangChain, Tensorflow, Pandas, Scikit-Learn, VGG16, ViT, Multi-View, Multi-Label
Frameworks: Blade, React, Angular, Vue.js, Django, Flask, Node.js, Spring Boot, Hadoop, Spark
Database: MySQL, Redis, MongoDB, PostgreSQL, DynamoDB, Oracle, Firebase, RocketMQ, Elasticsearch, MilvusDB
Tools: Git, Docker, AWS, Azure, CMake, Postman, CI/CD, Jenkins, Nginx, FFmpeg, OpenCV, Jira, Figma

PUBLICATION

• First Author: “Federated Multi-View Multi-Label Classification.” <i>IEEE Transactions on Big Data</i> , 2025.
• Co-Author: “Susceptibility genes identification and risk evaluation model construction by transcriptome-wide association analysis for salt sensitivity of blood pressure: the EpiSS study.” <i>BMC Genomics</i> , 2024.